



Herbal root extracts in Ben-Cha-Moon-Yai remedy attenuated pain-like behaviors and inflammation through the opioid and prostaglandin systems

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ABSTRACT

Ethnopharmacological relevance: Ben-Cha-Moon-Yai (BMY) remedy used in Thai traditional medicine as an anti-inflammatory, analgesic, and antipyretic agent comprises five herbal root extracts of equal weights: *Aegle marmelos* (L.) Corrêa (AM), *Oroxylum indicum* (L.) Kurz (OI), *Dimocarpus longan* Lour. (DL), *Dolichandrone serrulata* (Wall. ex DC.) Seem. (DS), and *Walsura trichostemon* Miq. (WT).

Aim of the study: To assess the anti-nociceptive and anti-inflammatory effects of the root extracts of all five species of BMY in experimental animal (mouse) models to ensure the rational use of herbal products in Thai traditional medicine.

Materials and methods: Root extracts prepared by ethanol and water extraction were used for the biological assays in animal models at five dose levels: 25, 50, 100, 200 & 400 mg/kg. The anti-nociceptive activity was evaluated based on hot-plate latency, duration of paw licking induced by formalin, and abdominal writhing induced by acetic acid. Carrageenan- and prostaglandin-induced paw oedema models were used to determine the anti-inflammatory activity.

Results: The oral administration of AM, DS and WT root extracts displayed significant analgesic effects in the hot-plate test, both phases (early and late) of formalin test and acetic-acid induced writhing test at different dose levels. OI and DL only produced significant analgesia in the late phase of the formalin test and writhing test. The pretreatment of animals with the non-selective opioid receptor antagonist naloxone, reverse AM, DS and WT induced-antinociceptive activity. In both carrageenan and prostaglandin-induced paw oedema tests, all five herbal plant root extracts significantly reduced paw oedema at 3 h or more at different dose levels. Rotarod test results showed no effects of five herbal plant root extracts on the balance and the motor coordination at the highest dose level evaluated (400 mg/kg).

Conclusion: The root extracts of AM, DS, and WT possess both central and peripheral anti-nociceptive properties, while OI and DL possess only peripheral analgesic properties. All five root extracts own anti-inflammatory properties, which might be due to their activity on the prostaglandin system. Altogether these findings ensure the rational use of BMY remedy in Thai traditional medicine.

1. Introduction

Pain has been defined as a discomforting sensation arising with the

existing or forthcoming tissue damage or linked with such kind of tissue damage (Cheng and Rosenquist, 2018). Pain is a subjective sensation, commonly categorized based on its pathophysiology as nociceptive,

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Table 1

Ethnopharmacological summary of five herbal plants in the BMY remedy, the plant parts used, and their analgesic effect and biological activity related to pain.

Plant species	Plant family	Part used	Major compounds	Analgesic activity/biological activity associated with pain	References
<i>Aegle marmelos</i> (L.) Corrêa	Rutaceae	Alcoholic extract of the barks	Marmelosin, umbelliferone, and para-coumaric acid	Reduce pain hypersensitivity in vincristine-induced neuropathic pain	Gautam and Ramanathan (2019)
		Fruits	Aegeline	Reduces reserpine-induced pain	Singh et al. (2021)
		Leaves	N/A	Anti-inflammatory, antipyretic and analgesic activities	Arul et al. (2005)
<i>Oroxylum indicum</i> (L.) Kurz	Bignoniaceae	Stem barks	N/A	Exhibit antinociceptive and anti-inflammatory activities	Lalrinzuali et al. (2016)
		Root barks	N/A	Improved CFA-induced arthritis mice via increased endogenous antioxidant enzymes	Karnati et al. (2013)
		The fresh fruit pods	γ -Sitosterol, 2-cyclohexen-1-one, 2-methyl-, benzene ethanol, 4-hydroxy-, 3-hydroxy-2-methylbenzaldehyde, cyclobutane carboxylic acid, decyl ester	Anti-inflammatory activity in RAW 264.7 macrophage induced by LPS	Dunkhunthod et al. (2020)
<i>Dimocarpus longan</i> Lour.	Sapindaceae	Pericarps	Ellagic acid, epicatechin, and gallic acid	Antioxidant and anti-inflammatory activity	Huang et al. (2012)
		Pulps	Chlorogenic acid, caffeic acid, coumarin, epicatechin, 4-ferulic acid, gallic acid, methyl catechol, quercetin, syringic acid, and vanillic acid	Antioxidant activity	Zhang et al. (2018)
<i>Dolichandrone serrulata</i> (Wall. ex DC.) Seem.	Bignoniaceae	Seeds	2"-O-apiosylverbascoside, decaffeoyl-verbascoside, dolichandroside, isoverbascoside, ixoside, luteoside, markhamioside A,B and verbascoside	N/A	Sinaphet et al. (2006)
		Flowers	Allerone, cleroidicin B, protocatechuic acid, ixoside, rengyolone, and isomaltose	Antioxidant activity	Phanthong et al. (2015)
<i>Walsura trichostemon</i> Miq.	Meliaceae	Roots	Trichostemonate	N/A	Sichaem et al. (2012)
		Twigs	α -Mangostin, trichostemonol, β -sitosterol, melianone, mangostenone	N/A	Sichaem et al. (2014)
		Stem barks	α -Mangostin, grandifolinolenone, Trichostemonate, 11 α ,20-dihydroxydammar-24-ene-3-one, sapelin E acetate	N/A	Phontree et al. (2014)
		Stems	Apotirucallane triterpenoids, dammarane, limonoid, 11,25-dideacetyltrichostemonate, 12 β , 20S, 24R-trihydroxydammar-25-en-3-one and 12 β , 20S, 25-trihydroxydammar-23-en-3-one.	N/A	Nugroho et al. (2021)

inflammatory, and neuropathic pain. Though pain is a common and essential sensation for the survival of human beings, it is even identified as the fifth vital sign in the healthcare system (Tompkins et al., 2017). The presence of pain is also linked with the increase in direct healthcare costs and associated indirect costs. Although non-opioid (acetaminophen, nonsteroidal anti-inflammatory drugs, tricyclic antidepressants) and opioid (codeine, morphine) analgesics are the mainstay of treatment in various types of pain, these drugs are associated with numerous side effects and low treatment efficacy. Therefore, many researchers are now interested in evaluating the anti-nociceptive effects of natural products to discover new analgesic drugs with higher efficacy and lower side effects profile. In this regard, numerous research has been conducted using medicinal plants used in folk medicine to evaluate their preclinical analgesic and anti-inflammatory efficacy. The potential analgesic effects are scientifically proven for plants, including *Morinda citrifolia* L., *Ricinus communis* L. (Sreekeesoon and Mahomoodally, 2014), *Aloe ferox* Mill., *Colocasia antiquorum* Schott, *Pycnostachys reticulata* (E. Mey.) Benth, *Senecio serratuloides* DC., and *Turraea floribunda* Hochst (Fawole et al., 2010)."

Ben-Cha-Moon-Yai (BMY) remedy is a commonly used Thai traditional medicine (TTM) (Fine Arts Department, 1999), comprises of five herbal root extracts in equal weights: *Aegle marmelos* (L.) Correa ex Roxb. (Rutaceae), *Oroxylum indicum* (L.) Kurz (Bignoniaceae), *Dimocarpus longan* Lour. subsp. *longan* var. *longan* (Sapindaceae), *Dolichandrone serrulata* (DC.) Seem. (Bignoniaceae), and *Walsura trichostemon* Miq. (Meliaceae) (Manohan et al., 2013). This remedy has been frequently used in TTM as an anti-inflammatory, antipyretic and analgesic agent. Many Thai traditional medicine books documented the ethnopharmacological uses of BMY remedy where it was reported to be used to treat "abnormal elements including wind and water, abscess poisoning, sputum, deformed tendon (or ligament) and muscle system" (Development of Applied Thai Traditional Medical Education

Foundation and Ayurved Thamrong School, 2016; Fine Arts Department, 1999). The cause of pain due to these abnormal elements, deformed tendon and muscle system has been documented in the Thai traditional medicine book (Development of Applied Thai Traditional Medical Education Foundation and Ayurved Thamrong School, 2020). Owing to the traditional use of this remedy, previous studies identifying the anti-nociceptive, anti-inflammatory (Kiratipaiboon et al., 2012) and antipyretic (Bansuttee et al., 2010) effects of the BMY remedy have been carried out to validate the folkloric uses. Apart from that, many research has been conducted to evaluate the biological activity of different components of the above five plants where the extracts of *A. marmelos* (Arul et al., 2005; Shankarananth et al., 2007), *O. indicum* (Islam et al., 2014; Lalrinzuali et al., 2016; Siriwatanametanon et al., 2009), *D. longan*, and *D. serrulata* (Ho et al., 2007) showed analgesic and anti-inflammatory properties. Further, the methods of standardizing the five herbal plant root materials in BMY remedy have been established to ensure the purity, quality and identification of root species (Manohan et al., 2013).

In this study, an attempt has been made to investigate the anti-nociceptive and anti-inflammatory effects of the root extracts of all five species of BMY in experimental animal models. Though our previous study has shown the anti-nociceptive efficacy of the BMY remedy, it would be beneficial to evaluate the antinociceptive and anti-inflammatory effects of five root extracts in BMY to ensure the rational use of herbal products by Thai traditional medicine practitioners.

2. Materials and methods

2.1. Ethnopharmacological search

Ethnopharmacological search on the five medicinal plants, *Aegle*

marmelos (L.) Corrêa (AM), *Oroxylum indicum* (L.) Kurz (OI), *Dimocarpus longan* Lour. (DL), *Dolichandrone serrulata* (Wall. ex DC.) Seem. (DS), and *Walsura trichostemon* Miq. (WT) was conducted using different literature sources to identify the analgesic, anti-inflammatory and antioxidant properties of the plants (Table 1).

2.2. Plant materials and chemicals

The roots of *A. marmelos*, *O. indicum*, *D. longan*, *D. serrulata*, and *W. trichostemon* were gathered from Chiangrai, Tak, Nakhon Ratchasima and Surin provinces, Thailand. The crudes were authenticated by Associate Professor Nijisiri Ruangrunsi (Manohan et al., 2013). The voucher specimens of *A. marmelos* (MAN_AM_04), *O. indicum* (MAN_OI_06), *D. longan* (MAN-DL_02), *D. serrulata* (MAN_DS_10), and *W. trichostemon* (MAN_WI_10) were stored at the College of Public Health Sciences, Chulalongkorn University, Thailand.

Indomethacin (IND) and morphine (MO), standard analgesic drugs and naloxone (NAL), were purchased from Sigma Chemical Co. (USA) and Thai FDA, respectively. Formalin and acetic acid were purchased from Merck, Germany. Normal saline solution (NSS) was provided by General Hospital Products Public Co., Thailand. 2% Tween 80 was purchased from Srichansahaosoth Co., Thailand.

2.3. Herbal root extraction

The root extracts and the remedy were prepared according to the same method described by Rawiwan (Manohan et al., 2013). Briefly, the roots were washed, air-dried under shade and ground to coarse powders. The dried-root powder was exhaustively macerated with absolute ethanol followed by water until exhaustion. Then the filtrates were evaporated (ethanolic extract) or lyophilized (water extract) to dryness. Obtained two fractions: ethanolic extract and water extract were weighed and stored at -20°C until use.

2.4. Pharmacological treatment

Various doses of AM, OI, DL, DS, WT (25, 50, 100, 200 and 400 mg/kg) and indomethacin (10 mg/kg) were prepared by suspending in 2% Tween 80. Morphine sulfate (10 mg/kg) and naloxone (5 mg/kg) were prepared by dissolving in NSS. The vehicle, 2% Tween 80 and the different concentrations of five root extracts were administered orally, and NSS (10 mL/kg) and MO (10 mg/kg) were administered intraperitoneally.

2.5. Experimental animals

Male ICR mice weighing 18–25 kg were purchased from the National Laboratory Animal Centre of Mahidol University, Thailand. All animals were maintained under properly controlled environmental conditions: temperature of $25 (\pm 2)^{\circ}\text{C}$, 50–60% humidity and 12 h light – 12 h dark cycles with a standard pellet diet and tap water *ad libitum*. Animals were allowed to acclimate to the animal facility for one week before the commencement of experiments. All experimental protocols were approved by the Institutional Animal Care and Use Committee, Faculty of Pharmaceutical Sciences, Chulalongkorn University (protocol number: 12-33-007).

2.6. Anti-nociceptive properties

2.6.1. Hot-plate analgesic test

Thermal pain response in animals was evaluated by using the hot-plate method (Woolfe and Macdonald, 1944). Briefly, the mice were placed separately on the hot-plate surface (Harvard Apparatus, USA) maintained at $55 \pm 0.5^{\circ}\text{C}$. The response latencies (licking of the hind paw or jumping up from the surface) were measured for seven subsequent trials at different time intervals for 4 h (15, 30, 45, 60, 90, 120,

and 240 min) after administration of controls or treatments. The 45 s was taken as the cut off value to avoid tissue injury. The time-course of heat latency was stated as the mean of percentage maximum possible effect (%MPE) and calculated as:

$$\%MPE = \frac{(\text{Postdrug latency}) - (\text{Predrug latency})}{(\text{Cutoff time}) - (\text{Predrug latency})} \times 100$$

The area of analgesia was determined by calculating the area under the time course vs %MPE curves applying the trapezoidal rule described by Tallarida (Gomm et al., 2011).

2.6.2. Writhing test

The acetic acid-induced writhing test was performed according to the method described by Koster (Koster et al., 1959) as a chemical pain model. In brief, 1 h after administering controls and treatments, 10 mL/kg of 0.6% acetic acid (v/v in saline) was injected intraperitoneally. Then the number of writhing movements were counted at 5 min intervals for 30 min. The anti-nociceptive effect was presented as percentage inhibition of writhing response and calculated as;

$$\% \text{ inhibition} = \frac{\text{Writhing (control)} - \text{Writhing (test)}}{\text{Writhing (control)}} \times 100$$

In this equation, “Writhing” means the average number of writhes.

2.6.3. Formalin test

The method described by Hunskaar in 1987 (Hunskaar and Hole, 1987) was used to evaluate the hind paw licking behavior induced by formalin in mice. In short, mice were injected with 20 μL of 2.5% formalin solution subcutaneously (to the left hind paw) 1 h after treating animals with different concentrations of five herbal root extracts and controls. Animals were immediately transferred into an observation cylinder. The time spent licking the injected hind paw was recorded in both the early phase (0–5 min) and the late phase (15–30 min) separately. The percentage inhibition of paw licking was calculated using the following formula.

$$\% \text{ inhibition of paw licking} = \frac{\text{Time (control)} - \text{Time (test)}}{\text{Time (control)}} \times 100$$

In the equation, time is the average time spent in paw licking (seconds).

2.6.4. Involvement of opioid system in the nociception of plant extracts

The possible contribution of the opioid system in the anti-nociceptive effects of herbal root extracts was examined using the previously described mouse hot-plate test. Mice were pre-treated with a non-selective opioid receptor antagonist, naloxone (5 mg/kg, i.p) 10 min before the oral administration of *A. marmelos* root extract, *D. serrulata* root extract or *W. trichostemon* root extract and the stimulation latencies were measured sequentially at 15, 30, 45, 60, 90, 120 and 240 min.

2.7. Anti-inflammatory properties

2.7.1. Carrageen-induced paw oedema test

The anti-inflammatory effect of different doses of five herbal root extracts of BMY was evaluated using paw oedema assay described by Winter (Winter et al., 1962) with minor alterations. Mice were treated with herbal extracts and control 1 h before the administration of the edematogenic agent. Oedema in the left hind paw was aided by subcutaneous injection of 50 μL of 1% carrageenan solution into the plantar surface of the paw. The paw volume was quantified using a plethysmometer (Ugo Basile, Italy) before and after carrageenan induction. The data are conveyed as mean variation in paw volume (mL) between the basal volumes and the volumes obtained after 1, 2, 3, 4, 5 and 6 h after administration of 1% carrageenan. Percentage inhibition of paw oedema was calculated as,

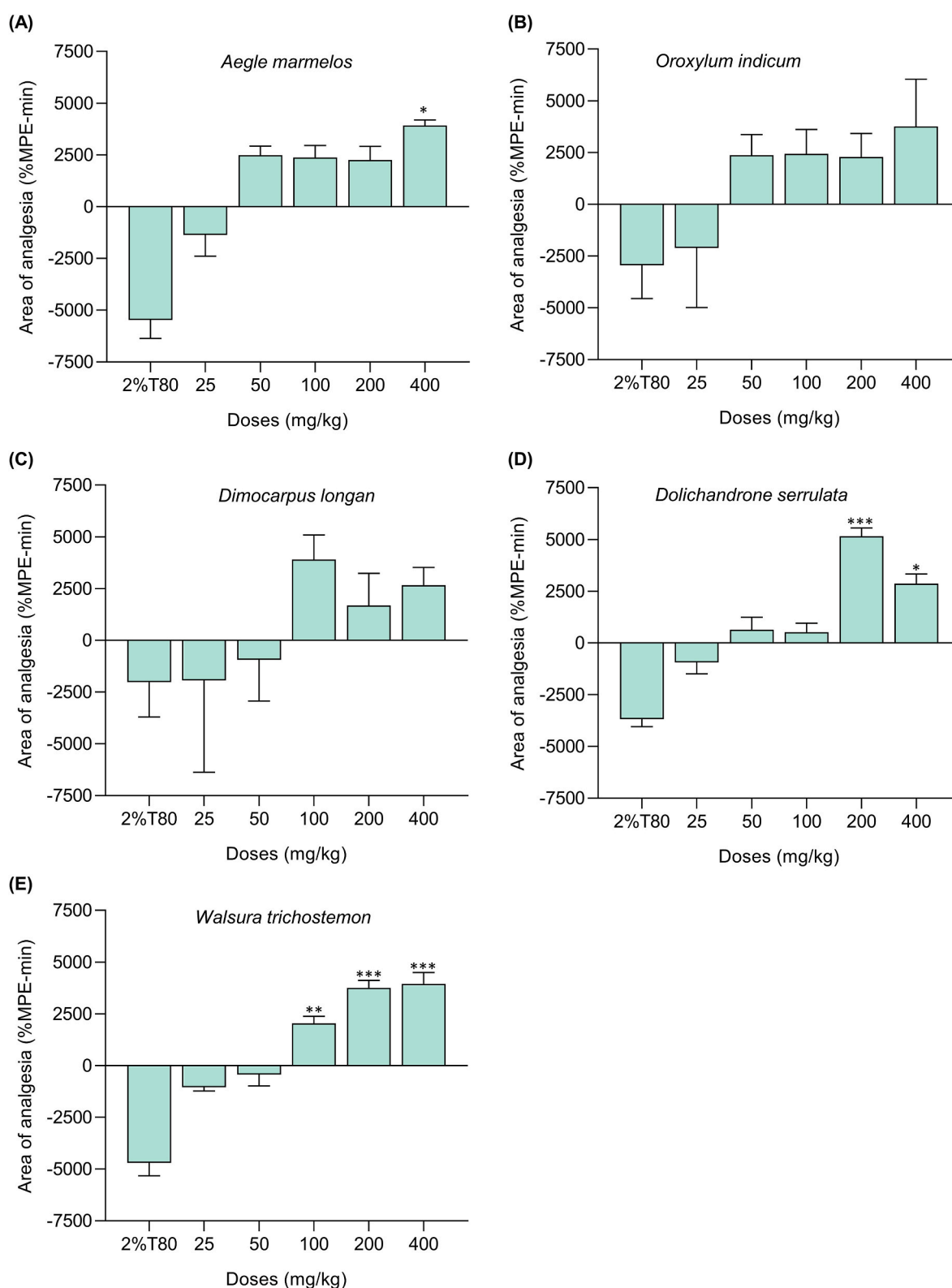


Fig. 1. Effects of five herbal plant root extracts on thermal nociception in mice. The thermal nociception was expressed as the area of analgesia (the AUC of % MPE-time curve for 0–240 min) after administration of 2% Tween 80 (2% T 80) and various doses (25–400 mg/kg) of different herbal plant root extracts in BMY remedy: n = 10 mice/group. Different graphs represent the effects of *Aegle marmelos* (A), *Oroxylum indicum* (B), *Dimocarpus longan* (C), *Dolichandrone serrulata* (D), and *Walsura trichostemon* (E), respectively. The *, ** and *** indicate the statistical significance compared to the vehicle group at the level of $p < 0.05$, $p < 0.01$, and $p < 0.001$, accordingly determined using ANOVA followed by the Tukey's post-hoc test.

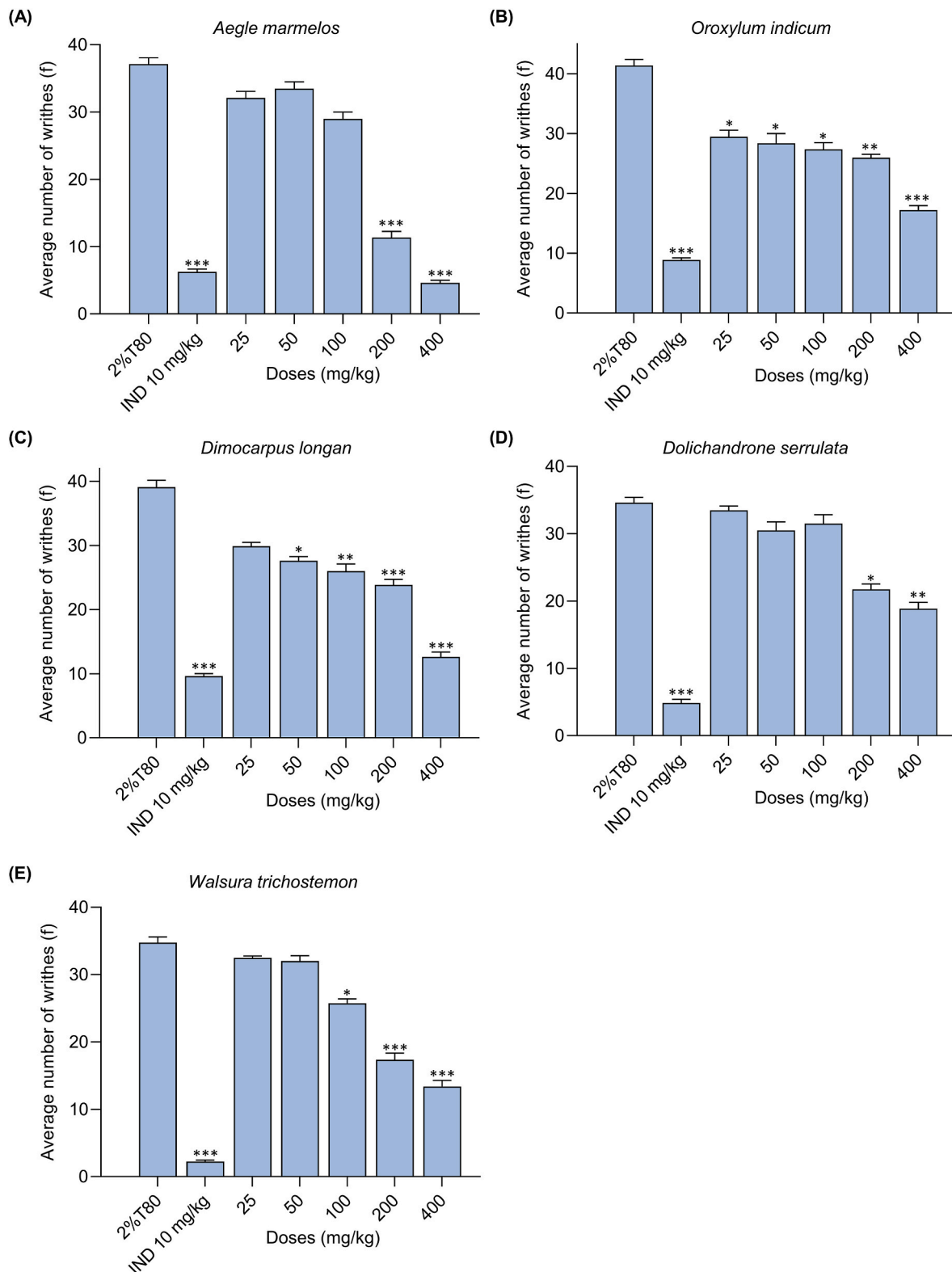


Fig. 2. Antinociceptive effects of different doses of five herbal plant root extracts, 2% Tween 80 (2% T80) and indomethacin 10 mg/kg in the acetic acid writhing test, (n = 8/group). The chemical nociception was expressed as the average number of writhes (measured for 30 min) in five different groups of *Aegle marmelos* (A), *Oroxyllum indicum* (B), *Dimocarpus longan* (C), *Dolichandrone serrulata* (D), and *Walsura trichostemon* (E). The *, **, and *** indicate the statistical significance compared to the vehicle group (Tween 80) at the level of $p < 0.05$, $p < 0.01$ and $p < 0.001$, respectively.

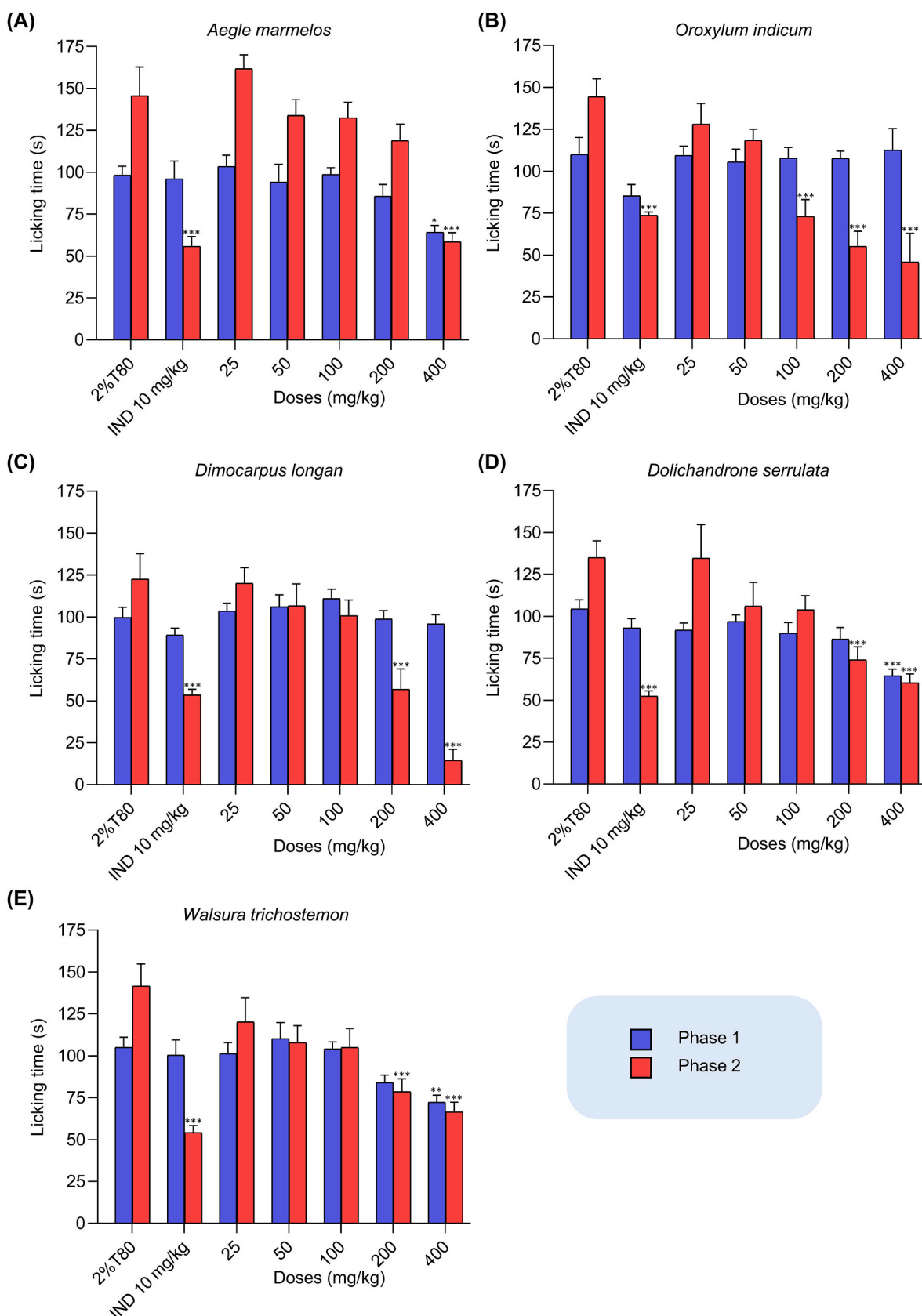


Fig. 3. Effects of five herbal plant root extracts on the early (blue bars) and late phases (red bars) of formalin test. Each graph represents the paw licking time in seconds after administration of 2% Tween 80 (2% T 80) and various doses (25–400 mg/kg) of different herbal plant root extracts in BMY remedy: n = 8/group. Graphs A, B, C, D, and E represent the effects of *Aegle marmelos*, *Oroxyllum indicum*, *Dimocarpus longan*, *Dolichandrone serrulata*, and *Walsura trichostemon* (WT), respectively. The *, ** and *** indicate the statistical significance compared to the vehicle group at the level of $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

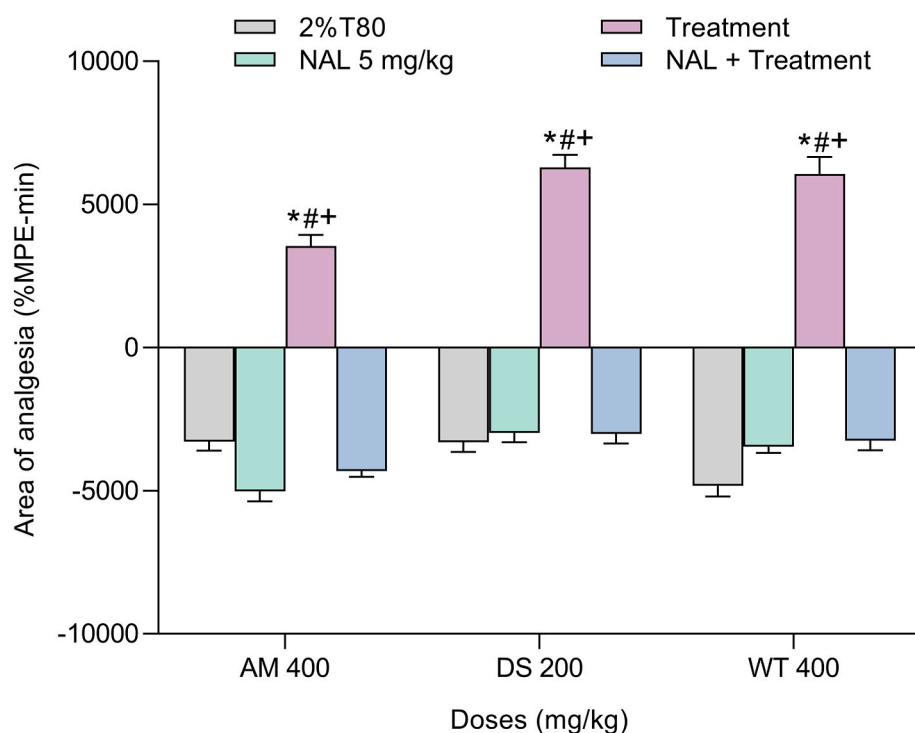


Fig. 4. Participation of the opioid system in the anti-nociception. The thermal nociception was expressed as the area of analgesia (the AUC of % MPE-time curve for 0–240 min) after administration of 2% Tween 80, naloxone alone, three herbal root extracts alone and with naloxone: $n = 10/\text{group}$. * indicates significantly different compared to 2% Tween 80 ($p < 0.001$), # indicates significant difference compared to naloxone ($p < 0.001$) and + indicates significant difference compared to the NAL+ treatment ($p < 0.001$); ANOVA followed by the Tukey's post-hoc analysis.

$$\% \text{ inhibition} = \frac{(V_c - V_t)}{V_c} \times 100$$

where V_c = paw volume (control group) and V_t = paw volume (test group).

2.7.2. Evaluation of mechanism of anti-inflammatory activity

The mechanism of anti-inflammatory activity of the highest dose (400 mg/kg) of five herbal root extracts of BMY (AM, OI, DL, DS and WT) was investigated using the PGE₂-induced acute inflammation model described by Akkol (Akkol et al., 2009). 0.01% PGE₂ solution (50 μL , SC) was used for the induction of inflammation. Drug administration and data collection followed the same method described in 2.6.1, and paw volume was measured before and after injection of PEG₂ at 0.5, 1, 1.5, 2, 3, and 4 h.

2.8. Rotarod test

The effect of root extracts of five herbal plants on motor coordination was evaluated by using the rotarod test previously described by Dunham and Miya (1957). Animals were placed on the horizontal rotating rod (3.5 cm diameter, 16 rpm) of the treadmill apparatus (Ugo Basile, Italy). Mice capable of retaining on the rotating rod equal to or more than 60 s in three consecutive trials were used for the study. Animals were treated with the highest concentration (400 mg/kg) of five herbal root extracts or 2% Tween 80, and motor coordination was evaluated at 30, 60, 90, 120 and 240 min of drug administration. Finally, the time durations that the animals could retain on the rotor-rod for 60 s were recorded.

2.9. Acute toxicity testing

The acute toxicity test was performed for five herbal root extracts to evaluate the possible toxicities. Here, the mice who underwent the rotarod test (test groups and control group) were examined for any morbidity and mortality for 72 h.

2.10. Statistical analysis

The results are presented as mean $\pm$ S.E.M. Differences in the mean values between test and control groups were analyzed by one-way ANOVA followed by *post hoc* Tukey's multiple comparisons. The student's *t*-test was used to compare two groups. $P < 0.05$ was used as the criterion of statistical significance.

3. Results

3.1. Anti-nociceptive effects

3.1.1. Effect on thermal nociception – hot-plate test

Firstly, the hot-plate test was used in this study to assess the effects of five root extracts on thermally induced acute pain. The standard analgesic drug, morphine (10 mg/kg, i.p.), significantly increased the hot-plate latency demonstrating an area of analgesia at 20138.4 %MPE-min compared to the control (–5219.9 %MPE-min; Supporting information S1: Fig. S1). The analgesic efficacy of the five herbal root extracts of the BMY remedy was evaluated at five concentration levels (25, 50, 100, 200 and 400 mg/kg), and the results are indicated in Fig. 1. As illustrated in the figure, the highest dose of AM (400 mg/kg, Figs. 1A), 200 and 400 mg/kg of DS (Figs. 1D), 100, 200 and 400 mg/kg of WT (Fig. 1E) produced significantly increased thermal analgesia compared to the vehicle group. The analgesic peak effect of AM 400 mg/kg was reached within 120 min of administration. The peak effects of DS 200 and 400 mg/kg were observed at 120 and 90 min of oral administration, respectively. WT produced significant analgesia at the dose level of 100 mg/kg and beyond (200 and 400 mg/kg) with the peak analgesia reached within 120, 30 and 90 min after oral administration, respectively (Supporting information S2: Table S2-1-4). Though all doses of OI and DL considerably increased the hot-plate latency compared to the vehicle group, the effects were not significant at the 95% confidence level.

3.1.2. Effect on chemical nociception – writhing test

The analgesic effects of five root extracts were further evaluated

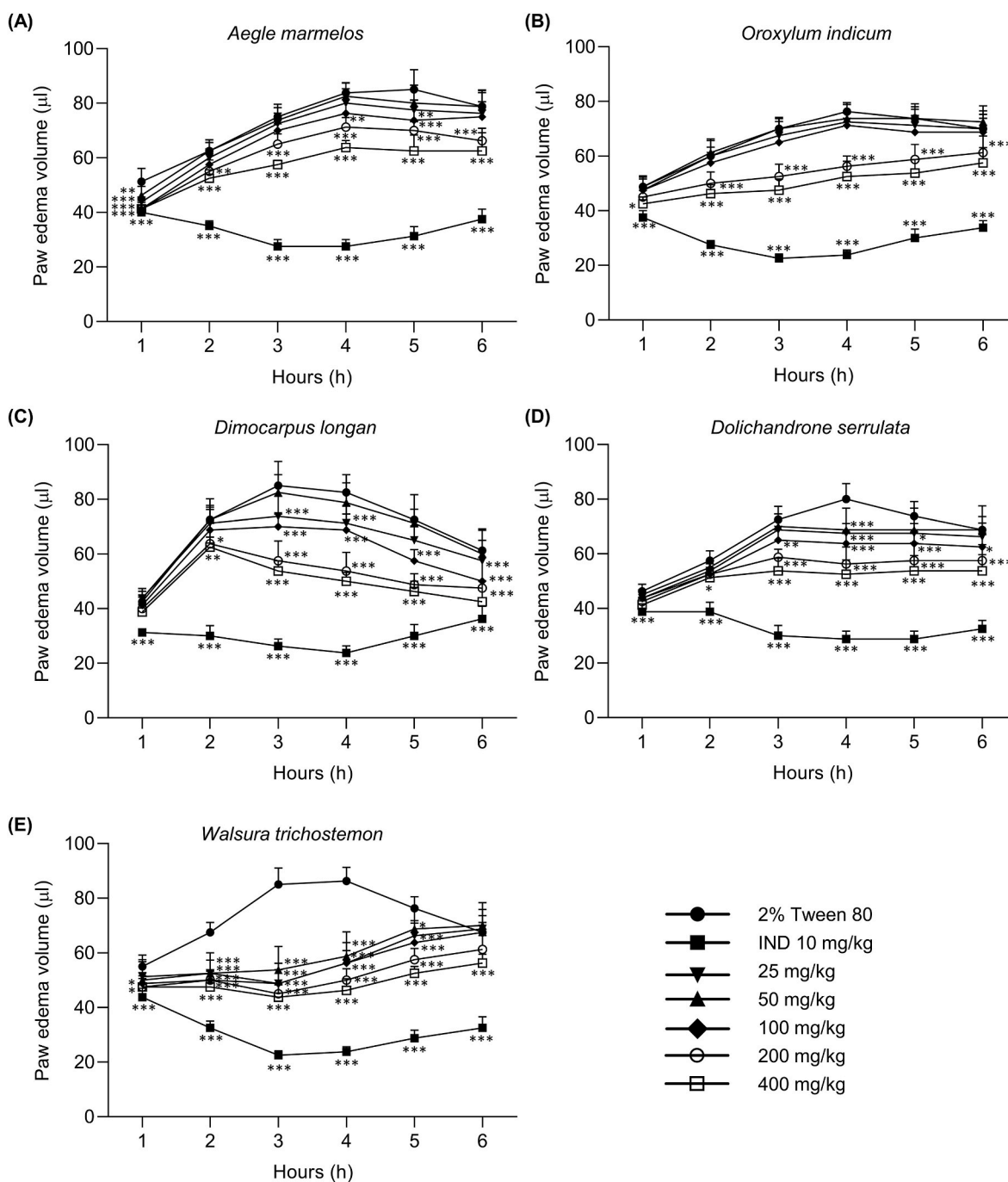


Fig. 5. Effects of five herbal plant root extracts on carrageenan-induced paw oedema in mice. Each graph represents the effects of 2% Tween 80, indomethacin 10 mg/kg and various doses of five herbal plant root extracts (A, B, C, D, and E graphs represent AM, OI, DL, DS, and WT, respectively) on the variation in paw volume (μL) obtained at 1–6 h after carrageenan administration compared to the basal paw volume ($n = 8/\text{group}$). The *, **, and *** indicate the statistical significance compared to the vehicle group (Tween 80) at the level of $p < 0.05$, $p < 0.01$ and $P < 0.001$, respectively.

against visceral pain using the acetic acid-induced writhing responses in mice. As shown in Fig. 2, indomethacin 10 mg/kg significantly reduced the writhing response (with the average inhibition of 83.31%) compared to 2% Tween 80 ($p < 0.001$). All five herbal plant root extracts showed a significant reduction in the average number of writhes induced by acetic acid at the dose levels of 200 and 400 mg/kg. Apart from that, 100 mg/kg dose of OI and DL showed 33.84% and 33.55% inhibitions of writhing response which are significantly different ($p < 0.05$) compared to the vehicle group. The anti-nociceptive efficacy of 200 and 400 mg/kg of AM (69.36% and 87.54% inhibition respectively) and 400 mg/kg of OI and DL (58.37% and 67.73% inhibition, respectively) are comparable to

10 mg/kg indomethacin.

3.1.3. Effect on formalin-induced pain

The hind paw licking behavior of mice evoked by formalin injection was used to evaluate the efficacy of root extracts on inflammatory pain. The formalin-induced nociception demonstrates a biphasic pattern: early phase (0–5 min after formalin injection), due to the direct stimulation of peripheral nociceptors and late phase (15–30 min after formalin injection) due to the response to inflammatory pain. According to the data collected, IND significantly decreased licking time only in the late phase compared with that of the vehicle group (Fig. 3). All five

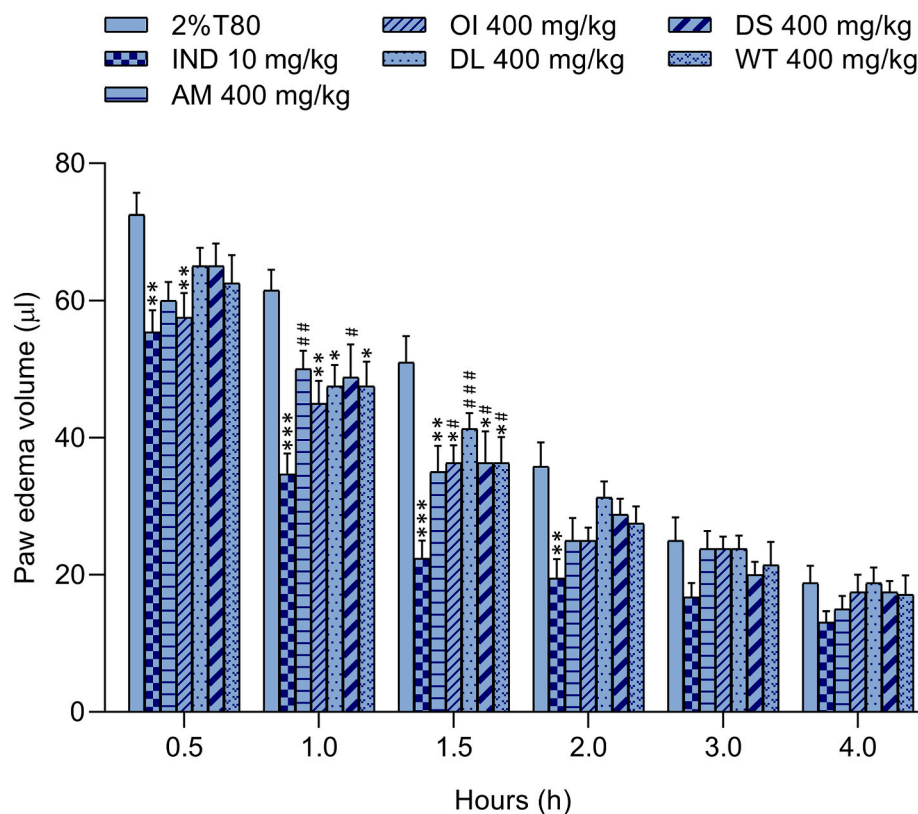


Fig. 6. Effects of five herbal plant root extracts (400 mg/kg) on PGE₂-induced inflammation in mice (n = 8/group). Each bar represents the effects of 2% Tween 80, indomethacin 10 mg/kg and five herbal plant root extracts (AM, OI, DL, DS, and WT, respectively) on the change in paw volume (μL) obtained at 0.5–4 h after PGE₂ administration compared to the basal paw volume (n = 8/group). The *, **, and *** indicate the statistical significance compared to the vehicle group (Tween 80) at the level of $p < 0.05$, $p < 0.01$ and $p < 0.001$, respectively. # $p < 0.05$, ## $p < 0.01$ and ### $p < 0.001$ vs. indomethacin group.

herbal plant root extracts showed a dose-dependent reduction in paw licking time in the late phase. AM, only at the dose level of 400 mg/kg, significantly decreased licking time in both the early phase (34.55% inhibition) and late phase (59.85% inhibition) compared with the vehicle group. All doses of OI and DL failed to decrease the licking time in the early phase. However, OI at the doses of 100, 200 and 400 mg/kg and DL at the doses of 200 and 400 mg/kg significantly decreased the licking time during the late phase by 49.40%, 61.78%, 68.28%, 53.57%, and 88.14% respectively compared to the vehicle group.

As same as AM, DS and WT produced a significant reduction in paw licking during the early phase by 38.17% and 31.66% compared to the vehicle group at the highest dose level (400 mg/kg), and a significant inhibition at the late phase was observed at the lower dose level of 200 mg/kg. 200 and 400 mg/kg of DS decreased the mean time spent on paw licking in the late phase by 45.22% and 55.31%, respectively, where the same doses of WT showed inhibition percentages of 44.49% and 53%, respectively, compared with the vehicle group.

3.1.4. Involvement of opioid system in the anti-nociceptive effects

The three plant extracts, which showed a significant analgesic effect and their concentration levels that showed the peak analgesia in the hot-plate test, were selected to evaluate analgesic mechanisms (AM and WT 400 mg/kg and DS 200 mg/kg). The possible participation of the opioid system in the anti-nociceptive effects of herbal root extracts assessed by the hot-plate test is shown in Fig. 4. As shown in the figure, all three root extracts (AM, DS, and WT) at the doses tested produce significant ($p < 0.001$) analgesic effect compared to the vehicle group. But that pain-relieving property of all three extracts was completely diminished after pretreatment with NAL (5 mg/kg), suggesting the involvement of the opioid system in the anti-nociceptive properties of the AM, DS, and WT root extracts.

3.2. Anti-inflammatory effects

3.2.1. Effect on carrageenan-induced paw oedema

In order to screen the anti-inflammatory effects of plant root extracts, carrageenan-induced paw oedema model in mice was used. According to the data obtained (Fig. 5), AM at the dose level of 400 mg/kg only produced a significant reduction in edematous response ($p < 0.05$) at 3 h and 4 h after carrageenan administration (23.33% and 23.93% inhibition, respectively). All doses of AM (25–400 mg/kg) produced lesser anti-inflammatory effects compared to IND 10 mg/kg. OI at the 200 mg/kg dose significantly decreased ($p < 0.05$) paw oedema at 3 and 4 h after administration with the percentage inhibition of 25% and 26.18%, respectively, compared to 2% Tween 80. OI at the 400 mg/kg dose produced a considerably higher anti-inflammatory effect than 200 mg/kg at 3, 4, and 5 h of oedema induction (65.77%, 56.45%, and 50% inhibition) are not comparable with indomethacin 10 mg/kg. Like OI, DL also significantly reduced paw oedema at the dose levels of 200 and 400 mg/kg at 3–5 h 400 mg/kg of OI showed maximum inhibition of paw oedema of 65.77% at 3 h after administering the edematous agent. A significant reduction of paw oedema was observed with 200 mg/kg of DS (29.69%) at 4 h and with 400 mg/kg of DS at 3 and 4 h (25.86% and 34.38% respectively) of carrageenan administration. Compared to the other four plant root extracts, WT showed considerable effectiveness in inflammation. 25–200 mg/kg of WT produced a significant anti-inflammatory response compared to 2% Tween 80 at 3 h and 5 h after induction. The highest dose of WT (400 mg/kg) showed significant anti-inflammatory effects from 3 to 5 h with 48.53% of maximum inhibition of oedema. The anti-inflammatory effects of all DL and DS and WT doses are not comparable with 10 mg/kg of IND ($p > 0.05$).

3.2.2. Involvement of PGE₂ on anti-inflammatory effects

The mechanism of five herbal root extracts of BMV remedy in producing anti-inflammation was evaluated using PGE₂-induced paw oedema model. The results are shown in Fig. 6. The figure shows that

indomethacin significantly inhibited the paw oedema on mice after 0.5–2 h after the stimulus. The pretreatment with 400 mg/kg of AM, OI, and DL significantly suppressed paw oedema at 0.5, 1 and 1.5 after the administration of PGE₂, showing the highest percentage inhibitions of 31.77%, 39.34% and 2.83%, respectively. The treatment with DS and WT significantly inhibited PGE₂ induced paw oedema only at 1–2 h after the stimulus. The highest percentages of inhibition produced by DS and WT were calculated as 27.5% and 29.34%, respectively, at 1.5 h after the induction of oedema.

3.3. Effect on motor-coordination – rotarod test

The rotarod test was used to assess the possible sedative and muscle relaxant effects of plant root extracts. Results showed no effects of five herbal plant root extracts at the highest dose level (400 mg/kg) on the balance and motor coordination. All five herbal plant root extracts did not modify the retention time (≈60 min) on the revolving rod at 30–240 min after administration (Supporting information S3).

3.4. Acute toxicity

The highest tested dose (400 mg/kg) of all five herbal root extracts had no effect on mice's morbidity and mortality during the observational period of 72 h after oral administration.

4. Discussion

The present study exemplified the contribution of five herbal plant root extracts of BMY remedy on its anti-nociceptive and anti-inflammatory actions. Although the anti-nociceptive and anti-inflammatory effects of different components of AM, OI, DL, and DS have been reported, there is not enough scientific data available for the biological activities of root extracts of those plants. Further, this is the very first study that demonstrated the analgesic and anti-inflammatory properties of WT.

The analgesic and anti-inflammatory effects of five herbal plant root extracts were evaluated using different animal models during this study. The oral route was used to administer root extracts into animals to resemble the usual route of BMY administration in Thai traditional medicinal practices. To estimate the specific analgesic effects of plant extracts, the hot-plate test, acetic acid-induced writhing test, and formalin test were used. Hence, this study infers the potential of plant root extracts in BMY remedy to exert therapeutic effects on distinct types of pain: acute thermal stimuli-induced acute pain, visceral pain, and inflammatory pain, respectively.

The hot-plate test is identified as a well-validated thermal nociception test, used to analyze central analgesic activity where the peripherally acting analgesics show negligible activity (Wang et al., 2015). In this study, the oral administration of AM, DS and WT root extracts displayed significant central analgesia in mice where the root extracts of OI and DL could not produce significant analgesia. These results show a contradiction to the hot-plate analgesic findings of *Oroxylum indicum* stem bark extract, where it showed significant central analgesic activity (Lalrinzuali et al., 2016). The mechanism of action of AM, DS, and WT appears to be, at least partially, derived from its effects on the opioid receptors as pretreatment of animals with the non-selective opioid receptor antagonist naloxone, reverted AM, DS and WT induced-antinociceptive activity.

To have a better understanding of the mechanism of analgesia of the above five plant root extracts, the formalin test was employed. Administration of formalin into rats or mice produces pain response in two distinct phases identified as early and late phases. Early phase pain response is mainly caused by the direct activation of C-fibre nociceptors where the prostaglandins do not play an important role. The late phase is predominantly ascribed to the inflammatory response induced by the formalin and functional alterations in the dorsal horn of the spinal cord

(Ezzatpanah et al., 2015). Centrally acting analgesics such as morphine inhibit both early and late phases evenly, whereas peripherally acting analgesics as aspirin only inhibit the late phase of formalin-induced pain (Hunskar and Hole, 1987; Rosland et al., 1990; Shibata et al., 1989). With relevance to this study, root extracts of AM, DS, and WT were able to produce significant analgesia in both phases of formalin-induced nociceptive pain at different dose levels suggesting central analgesic properties of these extracts. In contrast, root extracts of OI and DL produced significant analgesia only in the late phase suggesting the peripheral analgesic effects of those extracts. Moreover, these results are correlated with the findings of the hot-plate analgesic test.

To evaluate the peripheral analgesic effects of the root extracts, the acetic acid-induced writhing test was used. This method has been identified as the most useful animal model to evaluate analgesic effects due to its high selectivity (Wang et al., 2015). The administration of irritants such as acetic acid leads to the production and release of multiple pro-inflammatory mediators, including histamine, serotonin, bradykinin, prostaglandins and substance P, which evoke a writhing response in animals (Pingsusaen et al., 2015; Nakamura et al., 1986). In our study, all root extracts: AM, OI, DL, DS, and WT showed significant analgesic activity in this model, indicating the peripheral anti-nociceptive properties. These results are consistent with the findings of previous studies that reported analgesic properties of *Aegle marmelos* leaves extract (Arul et al., 2005; Shankarananth et al., 2007), *Oroxylum indicum* bark extract (Lalrinzuali et al., 2016; Zaveri and Sunita, 2010) and *Dimocarpus longan* seed extract (Ripa et al., 2014) in the acetic acid-induced writhing test.

The rotarod is a common test used to evaluate the substance's ability to explicit skeletal muscle relaxation, CNS depression, and convulsions (Garige et al., 2016; Stanley et al., 2005). The findings of this study reveal no evident sedative or relaxant effects in mice at the highest dose levels of all five plant root extracts. Hence, it can be declared that the behavioral responses evaluated in the above three animal models of pain were not likely due to the motor dysfunction but due to the anti-nociceptive effects of root extracts.

As there is an interconnection between the inflammatory response and nociception, the anti-inflammatory effects of five herbal root extracts of BMY remedy were evaluated using the carrageenan-induced paw oedema model. Immediately after the subcutaneous administration, carrageenan can induce acute, non-immune inflammatory response with cardinal signs: oedema, erythema, and hyperalgesia, mediated by several pro-inflammatory agents (Andrade et al., 2007; Kale et al., 2007). This test shows a biphasic response, where the first phase is mainly caused by the release of histamine and serotonin (1–2 h after the administration of carrageenan) and the second phase is mediated by prostaglandins where the kinins mediate the continuity between phases (Morris, 2003; DI ROSA, 1972). The findings of this study demonstrated that a significant reduction of paw oedema by indomethacin during the second phase, which can be explained by the fact that indomethacin is a cyclooxygenase inhibitor that contributes to the reduction of prostaglandin synthesis (Tsien, 1980). All five herbal plant root extracts significantly reduce paw oedema at 3 h or more at different dose levels, suggesting the anti-inflammatory effects in the second phase. Therefore, the anti-inflammatory effects of these five plant root extracts might be due to their effects on the release of prostaglandins or the blockage of prostaglandin receptors.

To identify the mechanism of anti-inflammatory activity, PGE₂ induced paw oedema model was used. PGE₂ is typically considered a crucial pro-inflammatory mediator, and its involvement in inflammation has enormously been studied. Elevated levels of PGE₂ have been detected in inflammatory exudates, and the direct tissue injection of PGE₂ has been shown to induce numerous classical signs of inflammation (Richardson and Vasko, 2002). In this study, the highest dose (400 mg/kg) of all five root extracts significantly inhibited oedema induced by PGE₂ at 0.5 h or more after PGE₂ administration, indicating the PGE₂ inhibitory effects of root extracts. Therefore, it is confirmed that all five

Table 2

Summary of the effective doses of the five root extracts of BMY remedy in various anti-nociceptive and anti-inflammatory models. (NS = no significant effect).

Root Extracts (mg/kg)	Anti-nociceptive				Anti-inflammatory	Reference
	Hot plate	Formalin test		Writhing test	Carrageenan-induced oedema	
		Early phase	Late phase			
AM	400	400	400	200,400	400	This study
OI	NS	NS	100,200,400	100,200,400	200,400	
DL	NS	NS	200,400	100,200,400	200,400	
DS	200,400	400	200,400	200,400	200,400	
WT	100,200,400	400	200,400	200,400	25,50,100,200,400	
BMY	125, 250, 500	250	125, 250, 500	125, 250, 500	125, 250, 500	Kiratipaiboon et al. (2012)

root extracts exert their anti-inflammatory action by blocking the effects of prostaglandins especially PGE₂. Further investigations are recommended to ascertain the discrete mechanism(s) of root extracts on inflammation.

Our previous study findings on anti-nociceptive and anti-inflammatory effects of BMY remedy published in 2012 (Kiratipaiboon et al., 2012) reveal the analgesic and anti-inflammatory activity of BMY remedy at all doses tested (125–500 mg/kg). As the BMY remedy used in Thai traditional medicine comprises five herbal root extracts of equal weights, 25–100 mg/kg of each extract forms 125–500 mg/kg of the remedy. When comparing the results (Table 2), the analgesic activity of the BMY remedy has more likely resulted from the effects of all five root extracts. The anti-inflammatory effect of the lowest dose of BMY remedy (125 mg/kg) is most likely from all five root extracts, while for higher doses of BMY (250–500 mg/kg), the effect seems to be mainly due to WT. But Thai traditional practitioners prefer to use the Ben-Cha-Moon-Yai remedy instead of single root extract as an anti-inflammatory or analgesic agent due to the belief that some herbal root extracts in remedy may reduce the toxic effects of other root extracts, which should be evaluated in future studies.

In the present study, the highest therapeutic dose of root extracts was evaluated in naïve mice. The results demonstrated no observable sedative effects and toxicity after 400 mg/kg administration of each extract in the rotarod and morbidity and mortality monitoring for 72 h, respectively. These results are in line with the previous studies, which showed the toxicity of the plant extracts at much higher doses compared to the maximum dose used in this study. Previous studies demonstrated that the leaves of *A. marmelos* Corr. extracted from alcohol, aqueous, methanolic exhibited toxicity at the dose of 1445, 1549, 1318 mg/kg, respectively, in a murine model of acute toxicity (Veerappan et al., 2007). Furthermore, a 5000 mg/kg dose of *Oroxylum indicum* ethanolic extract did not cause any toxic effects on the naïve mice (Reduan et al., 2020). Seed extract of *Dimocarpus longan* Lour. (longan) after oral administration showed LD₅₀ more than 5000 mg/kg in mice (Wor-asuttayangkurn et al., 2012). At the very high acute dose of fruit extracts of *Dimocarpus longan*, 20 g/kg, there was no behavioral toxicity, nor abnormal behaviors were observed (Chiranthanut et al., 2020). In addition, flower extracts of *Dolichandrone serrulate* showed no potential toxicity at a dose of 2000 mg/kg (Katisart and Konsue, 2019). All these results indicate that the dose of toxicity of the plants used in the Ben-Cha-Moon-Yai (BMY) remedy is far higher compared to its therapeutic doses.

Altogether, this study provides preclinical detail and comprehensive insight into five root extracts of BMY remedy as potential analgesic agents. Since BMY remedy has a traditional history of use or age-old usage, the safety and efficacy of the BMY for clinical application are more likely to be guaranteed. Hence, this study can potentially be translated into clinical trials. On that account, controlled human studies are warranted in future to prove the efficacy in patients with pain in the clinical setting.

5. Conclusion

The present study demonstrated that root extracts of AM, DS, and WT possess both central and peripheral anti-nociceptive properties, while OI and DL possess only peripheral analgesic properties. The analgesic mechanisms of AM, DS and WT are most likely involved with the opioid pathway. Apart from that, all five herbal root extracts demonstrated anti-inflammatory properties, which may involve the interference of the liberation of prostaglandins or inhibition of the effects of prostaglandin E₂. The mechanism of action of all five herbal root extracts in nociceptive and inflammatory pain should be further investigated, and more extensive toxicity studies are required to attest safety issues. These findings may eventually lead to the development of novel therapies with lesser adverse effects in the management of pain and inflammation and will provide additional scientific support to the Thai traditional medicinal system.

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Consent for publication

All the authors have given their consent to publish this paper.

Declarations of interest

None.

Availability of data and materials

The datasets used and analyzed during the current study are available from the corresponding author on reasonable request.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Chayanin Kiratipaiboon: Conceptualization, Methodology, Investigation, and, Formal analysis. **Peththa Wadu Dasuni Wasana:** Formal analysis, Writing – original draft, Writing – review & editing, and, Visualization. **Hasriadi:** Formal analysis, Writing – original draft, Writing – review & editing, and, Visualization. **Suchada Sukrong:** Writing – review & editing. **Nijsiri Ruangrungsri:** Resources, and, Writing – review & editing. **Pasarapa Towiwat:** Conceptualization, Methodology, Writing – review & editing, Supervision, and, Project administration.

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List of abbreviations

BMV	Ben-Cha-Moon-Yai
AM	<i>Aegle marmelos</i>
OI	<i>Oroxylum indicum</i>
DL	<i>Dimocarpus longan</i>
DS	<i>Dolichandrone serrulata</i>
WT	<i>Walsura trichostemon</i>
NSAIDs	non-steroidal anti-inflammatory drugs
TTM	Thai traditional medicine
TNF- α	tumour necrosis factor-alpha
IL-1 β	interleukin 1 beta
IL-6	interleukin-6
CFA	complete Freund's adjuvant
LPS	lipopolysaccharide
NO	nitric oxide
IND	indomethacin
MO	morphine
NAL	naloxone
NSS	normal saline solution
%MPE	percentage maximum possible effect
PGE ₂	prostaglandin E2
AUC	area under the curve

Appendix A. Supplementary data

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